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Peterson et al.

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(54) **FUEL CELL THROTTLE**

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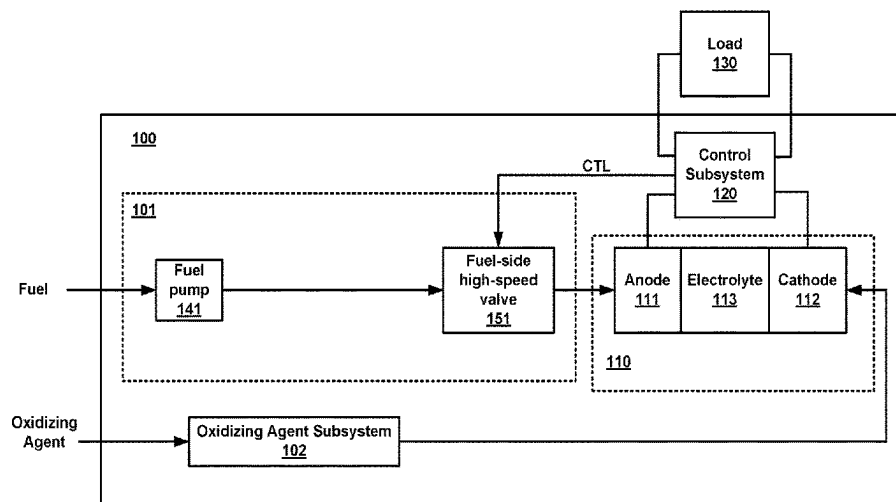
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(57) **ABSTRACT**

The disclosed technology is generally directed to fuel cells. In one example of the technology, a fuel cell stack that includes an anode and a cathode causes a load to be driven. A control subsystem is measures at least one characteristic associated with the load, and to provide at least one control signal based, at least in part, on the at least one character- istic. An oxidizing agent input subsystem provides an ox- idizing agent to the cathode of the fuel cell stack. A fuel input subsystem provides gaseous fuel to the anode of the fuel cell stack. The fuel input subsystem includes a fuel pump that is arranged to pump the gaseous fuel into the fuel input subsystem. A fuel-side high-speed valve adjusts mass flow of the gaseous fuel to the cathode of the fuel cell stack based on at least a first control signal of the at least one control signal.

13 Claims, 6 Drawing Sheets



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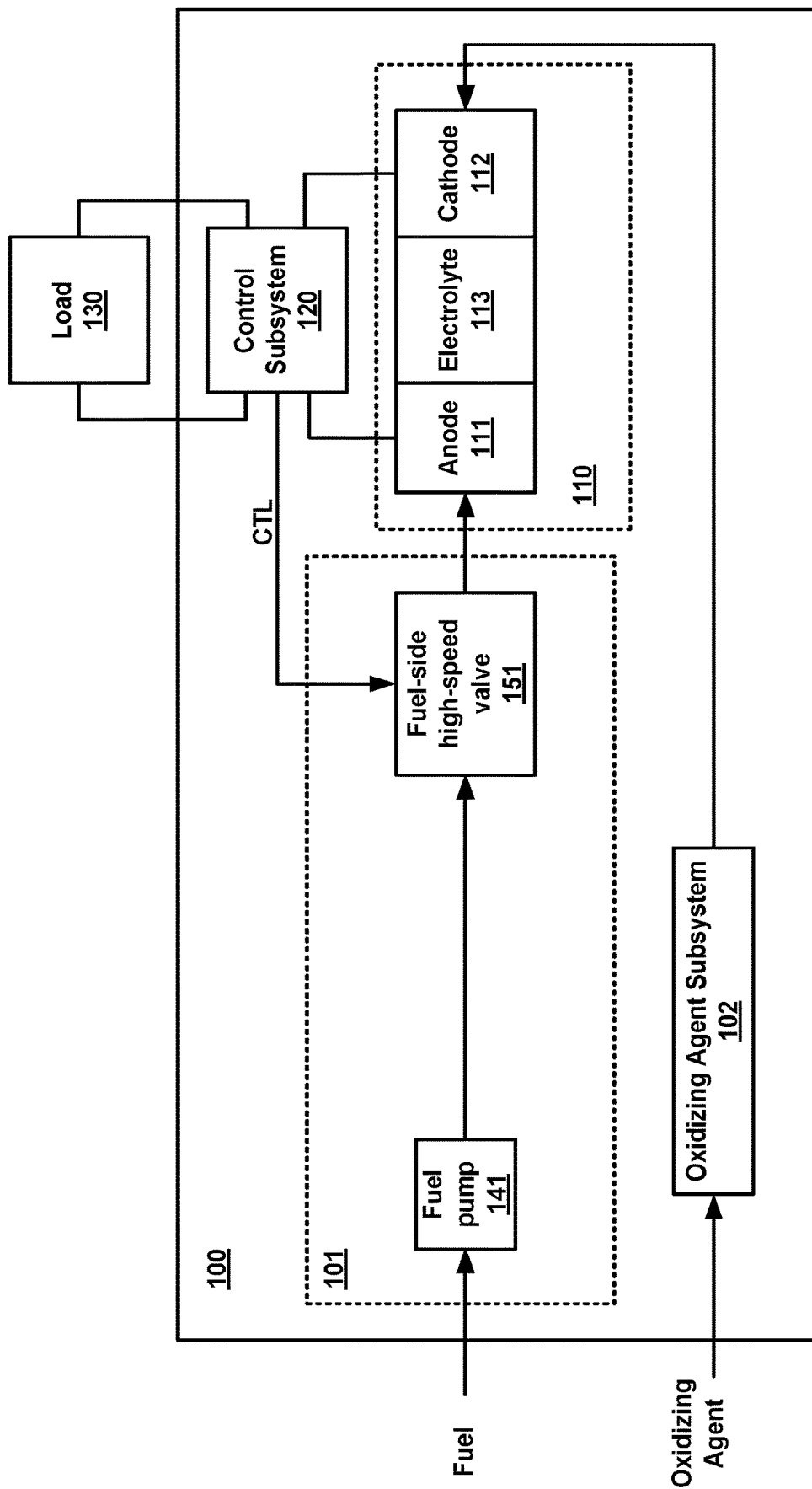


FIG. 1

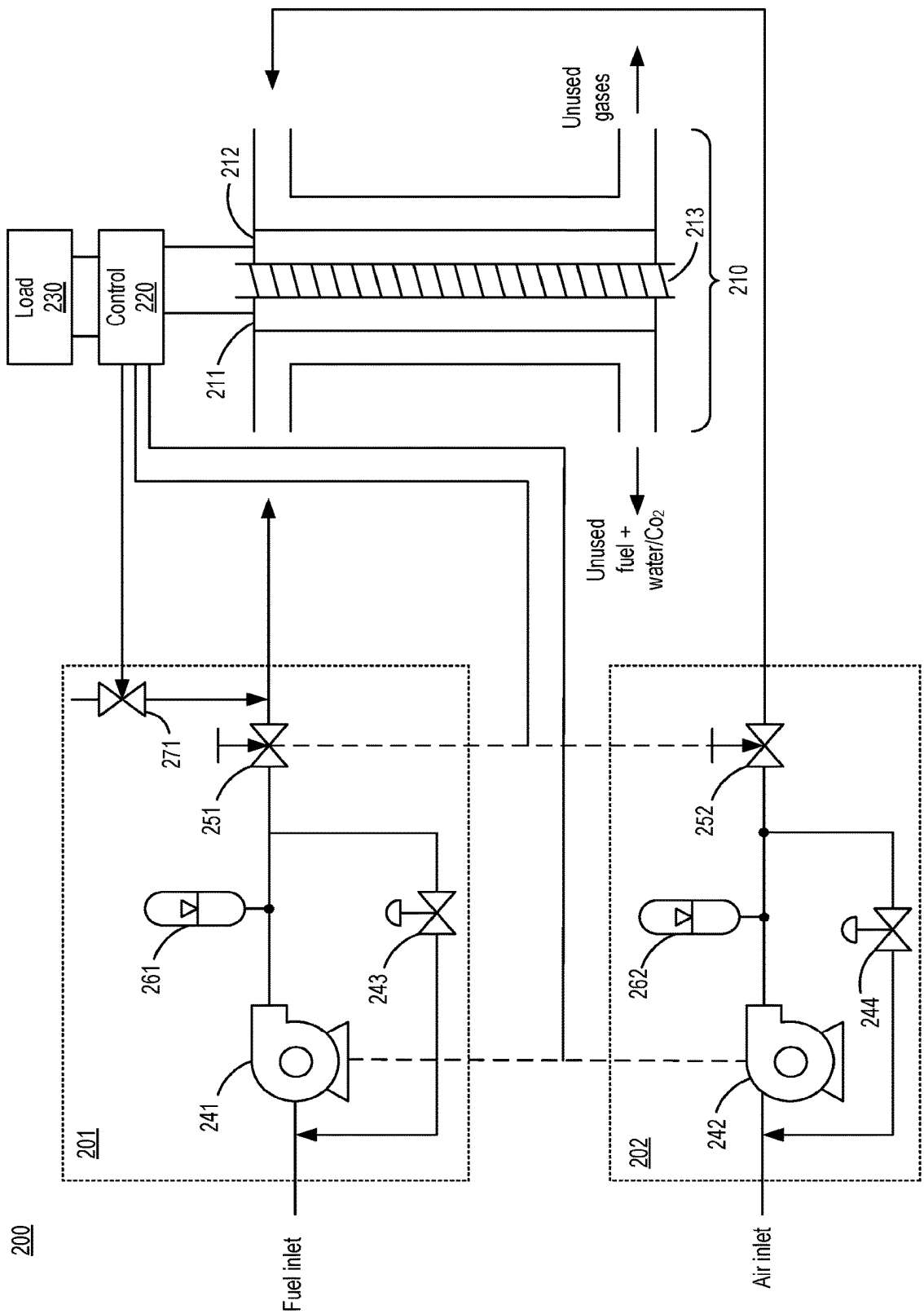


FIG. 2

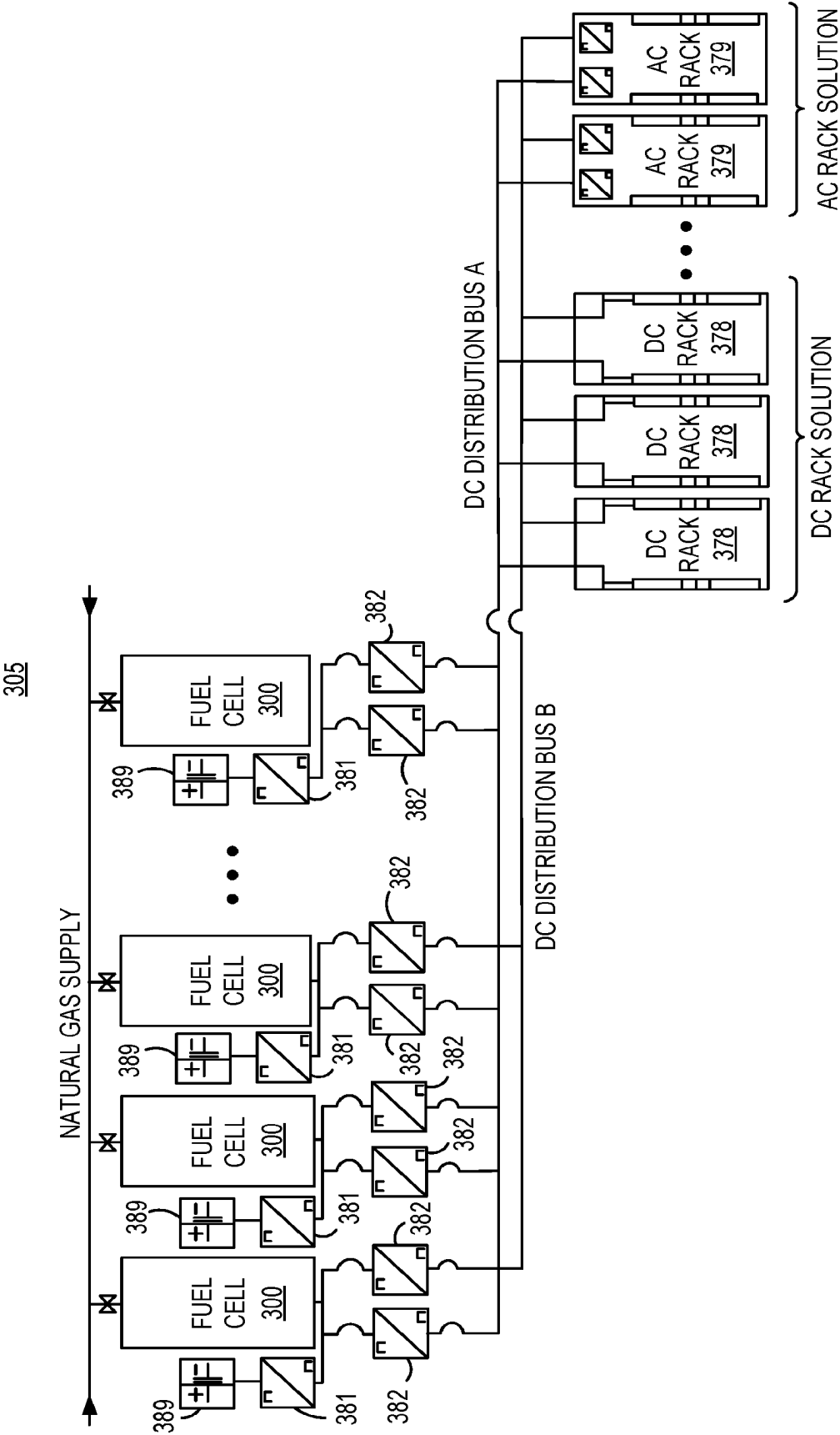


FIG. 3

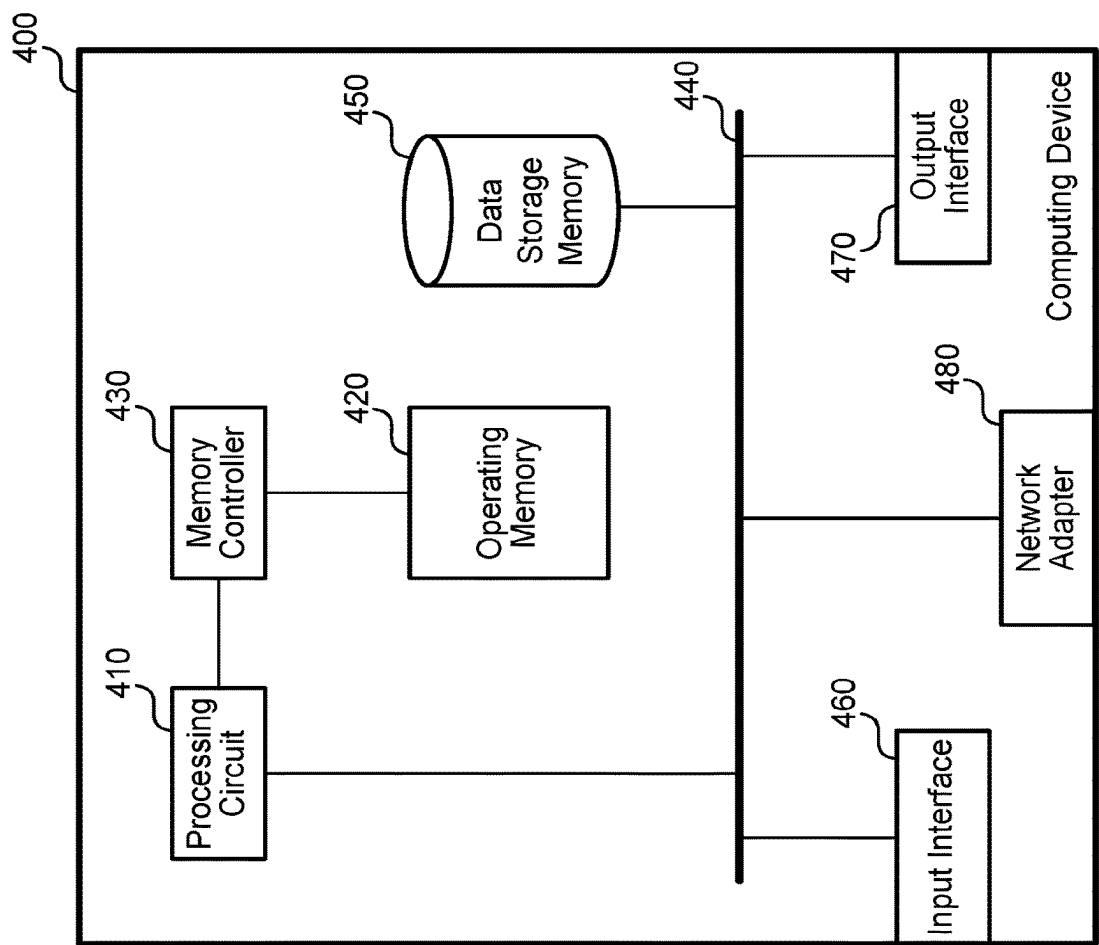


FIG. 4

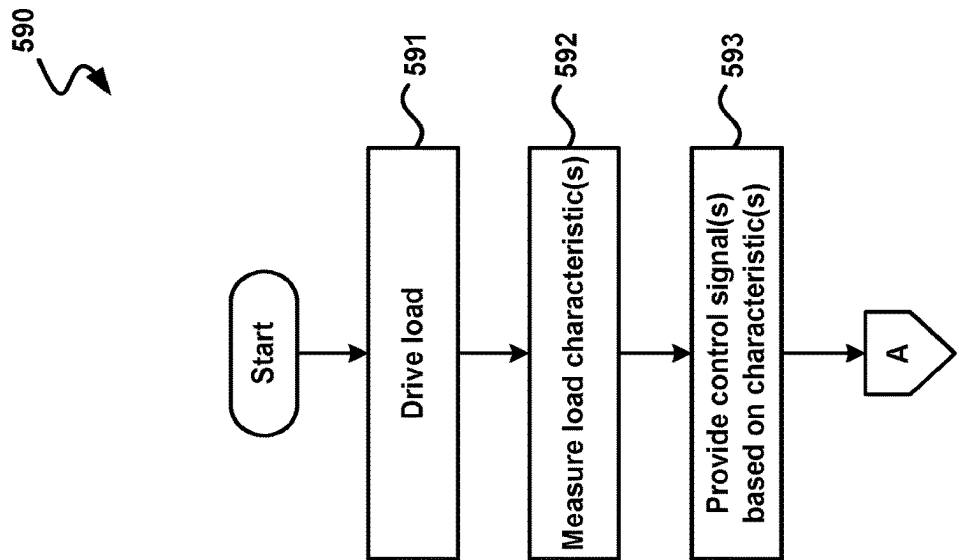


FIG. 5A

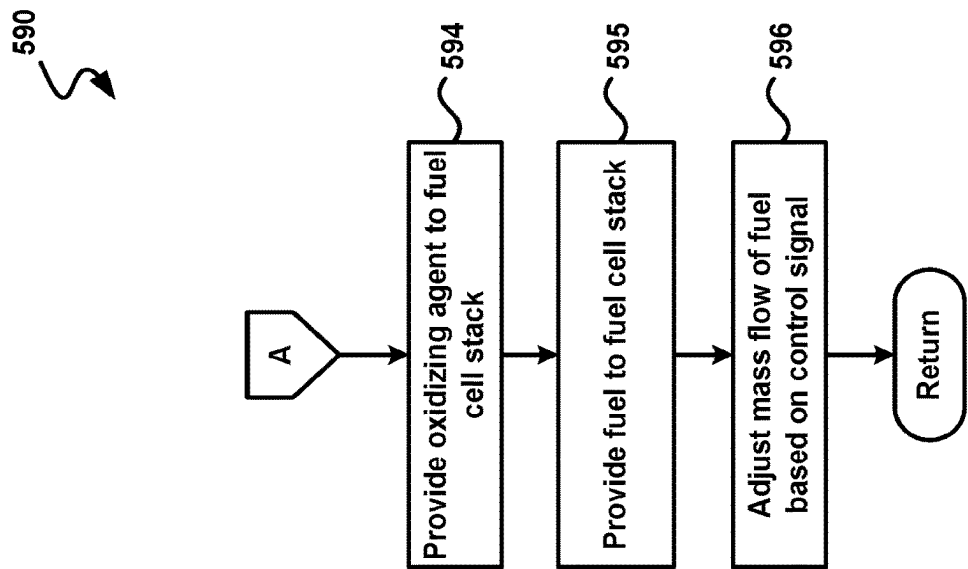


FIG. 5B

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FUEL CELL THROTTLE**CROSS-REFERENCE TO RELATED APPLICATION(S)**

This application is a continuation of U.S. patent application Ser. No. 16/429,488, filed Jun. 3, 2019, entitled “FUEL CELL THROTTLE”. The entirety of this afore-mentioned application is incorporated herein by reference.

BACKGROUND

Typically, fuel cells convert chemical energy from a fuel into electricity, by using an electrochemical process such as a chemical reaction of positively charged hydrogen ions or other fuel with oxygen or another oxidizing agent. Fuel cell operation typically depends on two supplies: a fuel supply which provides an ongoing source of fuel, and an oxidizing agent supply which provides an ongoing source of oxygen or other oxidizing agent to sustain the chemical reaction. The fuel is typically hydrogen, but other fuels may also be used. Various oxidizers may be used; in some cases, ambient air serves as an oxygen supply. Typically, unlike a battery, which eventually drains and must be recharged before further use as a power source, a fuel cell can produce electricity continuously as long as the necessary fuel and oxygen are supplied.

SUMMARY OF THE DISCLOSURE

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

Briefly stated, the disclosed technology is generally directed to fuel cells. In one example of the technology, a fuel cell stack includes an anode, a cathode, and an electrolyte that is situated between the anode and the cathode. In some examples, the fuel cell stack is configured to cause a load to be driven based, at least in part, on a voltage difference between the anode and the cathode. In some examples, a control subsystem is arranged to measure at least one characteristic associated with the load, and to provide at least one control signal based, at least in part, on the at least one characteristic. In some examples, an oxidizing agent input subsystem is arranged to provide an oxidizing agent to the cathode of the fuel cell stack. In some examples, a fuel input subsystem is arranged to provide gaseous fuel to the anode of the fuel cell stack. In some examples, the fuel input subsystem includes a fuel pump that is arranged to pump the gaseous fuel into the fuel input subsystem. In some examples, a fuel-side high-speed valve is arranged to adjust mass flow of the gaseous fuel to the cathode of the fuel cell stack based on at least a first control signal of the at least one control signal.

Other aspects of and applications for the disclosed technology will be appreciated upon reading and understanding the attached figures and description.

BRIEF DESCRIPTION OF THE DRAWINGS

Non-limiting and non-exhaustive examples of the present disclosure are described with reference to the following drawings. In the drawings, like reference numerals refer to

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like parts throughout the various figures unless otherwise specified. These drawings are not necessarily drawn to scale.

For a better understanding of the present disclosure, reference will be made to the following Detailed Description, which is to be read in association with the accompanying drawings, in which:

FIG. 1 is a block diagram illustrating an example fuel cell and load;

FIG. 2 is a block diagram illustrating an example of the fuel cell and load of FIG. 1;

FIG. 3 is a block diagram illustrating a portion of an example data center in which example fuel cells of FIG. 1 and/or FIG. 2 may be used;

FIG. 4 is a block diagram illustrating one example of a suitable computing device; and

FIGS. 5A-5B are a flow diagram illustrating an example of a process for fuel cell throttling, in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

The following description provides specific details for a thorough understanding of, and enabling description for, various examples of the technology. One skilled in the art will understand that the technology may be practiced without many of these details. In some instances, well-known structures and functions have not been shown or described in detail to avoid unnecessarily obscuring the description of examples of the technology. It is intended that the terminology used in this disclosure be interpreted in its broadest reasonable manner, even though it is being used in conjunction with a detailed description of certain examples of the technology. Although certain terms may be emphasized below, any terminology intended to be interpreted in any restricted manner will be overtly and specifically defined as such in this Detailed Description section. Throughout the specification and claims, the following terms take at least the meanings explicitly associated herein, unless the context dictates otherwise. The meanings identified below do not necessarily limit the terms, but merely provide illustrative examples for the terms. For example, each of the terms “based on” and “based upon” is not exclusive, and is equivalent to the term “based, at least in part, on”, and includes the option of being based on additional factors, some of which may not be described herein. As another example, the term “via” is not exclusive, and is equivalent to the term “via, at least in part”, and includes the option of being via additional factors, some of which may not be described herein. The meaning of “in” includes “in” and “on.” The phrase “in one embodiment,” or “in one example,” as used herein does not necessarily refer to the same embodiment or example, although it may. Use of particular textual numeric designators does not imply the existence of lesser-valued numerical designators. For example, reciting “a widget selected from the group consisting of a third foo and a fourth bar” would not itself imply that there are at least three foo, nor that there are at least four bar, elements. References in the singular are made merely for clarity of reading and include plural references unless plural references are specifically excluded. The term “or” is an inclusive “or” operator unless specifically indicated otherwise. For example, the phrases “A or B” means “A, B, or A and B.” As used herein, the terms “component” and “system” are intended to encompass hardware, software, or various combinations of hardware and software. Thus, for

example, a system or component may be a process, a process executing on a computing device, the computing device, or a portion thereof.

Briefly stated, the disclosed technology is generally directed to fuel cells. In one example of the technology, a fuel cell stack includes an anode, a cathode, and an electrolyte that is situated between the anode and the cathode. In some examples, the fuel cell stack is configured to cause a load to be driven based, at least in part, on a voltage difference between the anode and the cathode. In some examples, a control subsystem is arranged to measure at least one characteristic associated with the load, and to provide at least one control signal based, at least in part, on the at least one characteristic. In some examples, an oxidizing agent input subsystem is arranged to provide an oxidizing agent to the cathode of the fuel cell stack. In some examples, a fuel input subsystem is arranged to provide gaseous fuel to the anode of the fuel cell stack. In some examples, the fuel input subsystem includes a fuel pump that is arranged to pump the gaseous fuel into the fuel input subsystem. In some examples, a fuel-side high-speed valve is arranged to adjust mass flow of the gaseous fuel to the cathode of the fuel cell stack based on at least a first control signal of the at least one control signal.

In some examples, a fuel cell includes a pump to bring fuel into the system and a high-speed valve near the stack to adjust mass flow of the fuel into the stack based detected load changes. In some examples, this allows the fuel cell to follow load changes relatively quickly, such as on the order of milliseconds.

Illustrative Fuel Cell

FIG. 1 is a diagram of an example of fuel cell **100** and load **130**. Fuel cell **100** may be any suitable type of fuel cell. Fuel cell **100** may include fuel cell stack **110**, control subsystem **120**, fuel input subsystem **101**, and oxidizing agent input subsystem **102**. Fuel cell stack no may include anode in, cathode **112**, and electrolyte **113**. Fuel input subsystem **101** may include fuel pump **141** and fuel-side high-speed valve **151**.

In fuel cell stack **110**, in some examples, electrolyte **113** is situated between anode in and cathode **112**. Fuel cell stack no may be configured to cause load **130** to be driven based, at least in part, on a voltage difference between anode in and cathode **112**. In some examples, fuel cell stack no is in a hot box and is at a substantially higher temperature than the rest of fuel cell **100** during normal fuel cell operation. Fuel cell stack no may receive a fuel from fuel input subsystem **101** and an oxidizing agent from oxidizing agent subsystem **102**. The internal fuel may be, for example, hydrogen, carbon monoxide, or the like. In some examples, the external fuel provided may a fuel that reacts inside the hot box to provide the internal fuel—for instance, in some examples, the external fuel may be a hydrocarbon such as methane that reacts to form hydrogen and carbon monoxide, and in turn hydrogen and carbon monoxide act as the internal fuel.

The oxidizing agent may be, for example, oxygen or an oxide. The oxidizing agent may be a gas, such as air, that includes oxygen. The fuel cell stack may rely upon a chemical reaction, such as the combination of hydrogen and oxygen to form water, or the reaction of carbon monoxide and oxygen to form carbon dioxide.

Electrolyte **113** may allow ions to move between anode **111** and cathode **112** of fuel cell stack no. At anode in, a catalyst may cause the fuel to undergo oxidation reactions that generate ions and electrons. The ions may move from anode in to cathode **112** through electrolyte **113**. At the same time, electrons may flow from anode in to cathode **112**

through load **130**, producing a current. At cathode **112**, another catalyst may cause ions, electrons, and the oxidizing agent to react, forming outputs of the chemical reaction, such as water in the case of hydrogen and oxygen. During operation, cell stack no may provide a voltage drop of about 0.7 V between anode in and cathode **112**, where the voltage drop may vary based on the load and other factors.

Control subsystem **120** is arranged, in some examples, to measure at least one characteristic associated with load **130**, such as current, voltage, and/or the like, and to provide one or more control signals based, at least in part, on the measured characteristics. In some examples, a look-up table may be used to provide one or more of the control signals based on the measured characteristics.

In some examples, oxidizing agent input subsystem **102** is arranged to provide the oxidizing agent to cathode **112** of fuel cell stack no. In some examples, fuel input subsystem **101** is arranged to provide the fuel to cathode in of fuel cell stack no. In some examples, fuel pump **141** is arranged to pump the fuel into fuel input subsystem **101**, where the fuel is a gaseous fuel. In some examples, fuel-side high-speed valve **151** is arranged to adjust mass flow of the gaseous fuel to anode of the stack based on at least one of the control signals (e.g., control signal CTL) provided by control subsystem **120**. For instance, in some examples, fuel-side high speed valve **151** is controlled to adjust the mass flow of the fuel provided to fuel cell stack **110** based on the load current and load voltage, with those values provided to a look-up table, so that a change in load is followed relatively quickly, such as in a time frame of milliseconds.

FIG. 2 is a diagram of an example of fuel cell **200** and load **230**, which may be examples of fuel cell **100** and load **130** of FIG. 1. Fuel cell **200** may include fuel cell stack **210**, control subsystem **220**, fuel input subsystem **201**, and oxidizing agent input subsystem **202**. Fuel cell stack **210** may include anode **211**, cathode **212**, and electrolyte **213**. Fuel input subsystem **201** may include fuel pump **241**, pressure regulator **243**, fuel accumulator **261**, fuel-side high-speed valve **251**, and heated nitrogen purge subsystem **271**. Oxidizing agent input subsystem **202** may include oxidizing agent pump **242**, pressure regulator **244**, oxidizing agent accumulator **262**, and oxidizing-agent-side high-speed valve **252**.

In some examples, fuel pump **241** and oxidizing agent pump **242** are variable speed pumps, with the speed of fuel pump **241** linked with the speed of oxidizing agent pump **242**, with each being controlled based on one of the control signals provided by control subsystem **220**. In some examples, fuel pump **241** is arranged to increase the pressure of the fuel from the fuel inlet into fuel cell **200**, and oxidizing agent pump **242** is arranged to increase the pressure of the oxidizing agent from the oxidizing agent inlet into fuel cell **200**. Fuel input subsystem **201** may include one or more pipes through which the gaseous fuel passes, and oxidizing input subsystem **202** may include one or more pipes through which the oxidizing agent, such as air or a solid oxide, passes.

In some examples, pressure regulator **243** is arranged to regulate the pressure of the fuel into fuel cell **200**, and pressure regulator **244** is arranged to regulate the pressure of the oxidizing agent into fuel cell **200**. In some examples, pressure regulator **243** is part of a return line that bypasses around pump **241** to ensure that flow of fuel into fuel cell **200** is the correct volume and ratio. In some examples, pressure regulator **244** is part of a return line that bypasses around pump **242** to ensure that flow of oxidizing agent into fuel cell **200** is the correct volume and ratio. In some

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examples, pressure regulator **243** is part of fuel pump **241**, and in other examples, pressure regulator **243** is separate from fuel pump **241**. Similarly, in some examples, pressure regulator **244** is part of fuel pump **242**, and in other examples, pressure regulator **244** is separate from fuel pump **242**.

In some examples, fuel-side high-speed valve **251** is arranged to control mass flow of the fuel to fuel cell stack **210** based on control provided by control subsystem **220**. In some examples, oxidizing-agent-side high-speed valve **252** is arranged to control mass flow of the oxidizing agent to fuel cell stack **210** based on control provided by control subsystem **220**. The control provided to high-speed valves **251** and **252** may be provided by control subsystem **220** via feedback based on load **230**, such as by pulse width modulation (PWM) or other suitable type of feedback. In some examples, the control provided to high-speed valve **251** and high-speed valve **252** are mechanically and/or electrically linked. In this way, in some examples, high-speed valve **251** and high-speed valve **252** are synchronized with each other. One example of mechanical linkage would be one actuator controlling both high-speed valve **251** and high-speed valve **252**. High-speed valve **251** and/or high-speed valve **252** may each be a high-speed needle valve, a high-speed butterfly valve, or other suitable high-speed valve.

In some examples, the high-speed valves **251** and **252** are each close to fuel cell stack **210**. In some examples, each high-speed valve (**251** and **252**) includes an actuator that is outside of the hot box, along with a long shaft or plunger that goes inside the hot box, opening up into a valve inside the hot box, so that the opening of the high-speed valve is inside the hot box. The actuator may include electronics, a motor that turns the valve, and/or the like. Also, control associated with the high-speed valve may be outside of the hot box.

As discussed above, in some examples, the hot box contains the fuel cell stack **210**, which reacts at a high temperature, with the heat box being a well-insulated containing the high-temperature stack and providing insulative protection from high temperatures to the rest of the fuel cell, keeping the heat inside the hot box during operation of the fuel cell. In some examples, the electronics are kept outside of the hot box. In some examples, pumps **141** and **142** are also kept a certain distance away from the hot box to avoid having pumps **141** and **142** from being at high temperature during fuel cell operation. That is, in some examples, while high-speed valves **251** and **252** are quite close to the fuel cell stack **210**, almost right at the input of fuel cell stack **210**, pumps **241** and **242** are relatively far from fuel cell stack **210**.

Accumulator **261** may be arranged to handle pressure drops during the opening of valve **251** and the ramping up of pump **241** during high-speed load demand changes. Accumulator **261** stores fuel as high pressure. In this way, for example, if the pressure drops suddenly because the flow at valve **251** is open, accumulator **261** can provide additional pressure, or an additional buffer of fuel. In this way, in some examples, valve **251** and accumulator **261** work together to enable fuel cell **200** to adjust to rapid changes in the load—when valve **251** is open, a pressure drop may occur, and accumulator **261** may provide additional volume to ensure that the fuel can ramp up and compensate in time. Because the fuel is gaseous, in some examples, when the fuel is pressurized, the gas is compressed, so that it is possible to get more fuel mass through the pipe in fuel input subsystem **201**.

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Accumulator **262** may operate in a similar manner on the oxidizing agent side to provide an additional buffer of oxidizing agent, for example when the flow at valve **252** is open.

In some examples, a parallel fuel input may be used instead of or in addition to accumulator **261**. The parallel fuel input may provide a second path for fuel that joins the main path right near anode **211** of fuel cell stack **210**, that opens up and begins injecting higher pressure fuel there if the load increases very quickly, and a staged system may be used in some examples.

In some examples, heated nitrogen purge subsystem **271** may be used to “de-throttle” fuel cell **200** in the event of an instantaneous loss of load, so as to prevent excess fuel from passing through the system. In some examples, heated nitrogen purge subsystem **271** may also be used in the event of a sudden and significant drop in the load, with the amount of heated nitrogen provided based on the drop in load as detected by control subsystem **220**. In some examples, heated nitrogen purge subsystem **271** is used upon shutdown of fuel cell **100** to ensure that no further power is generated. Although FIG. **2** specifically shows a heated nitrogen purge subsystem, in other examples, a different purge subsystem may be employed, so that an inert gas other than nitrogen may be used for the purge.

As discussed above, control subsystem **220** may provide one or more control signals based on load current, load voltage, and/or the like. In some examples, other factors such as temperature of fuel cell stack **210**, volumetric measurements on the oxidizing agent side, differential pressure, fuel flows, and/or the like may also be used by control subsystem **220**, particularly for control of the oxidizing agent side. In some examples, pressure sensors may be used throughout the pipes on the fuel side and the oxidizing agent side and provide the output of the pressure sensors to control subsystem **220**. Control subsystem **220** may provide control to both the fuel side and the oxidizing-agent side. In some examples, control of the oxidizing-agent side may ensure that no large pressure differential develops between the fuel side and the oxidizing-agent side.

Fuel cell stack **210** may output unused fuel, unused oxidizing agent, and one or more outputs of the chemical reaction, such as, for example, water. Although not shown in FIG. **2**, rather than providing the voltage drop across anode **211** and cathode **212** directly to load **230**, in some examples, a power regulator is first used to ensure that the output DC voltage is relatively constant, and the constant DC voltage output is applied to the stack. In some examples, where output AC power is required, an AC/DC converter may be used to convert the power to AC power before being applied to the load. In some examples, the output voltages of multiple stacks from multiple fuel cells may be connected in series. For instance, in some examples, a number of fuel cells may each provide a voltage drop across the respective anode and diodes of the fuel cell stack, with the voltage drop provided to a voltage regulator to provide a substantially constant voltage, with the voltage output of each of the power regulators coupled together to provide a combined voltage output that is provided across a load. When the load changes, each of the fuel cells may then follow the load accordingly at a range of speed of milliseconds based on the variable-speed pumps, accumulators, and high-speed valves present in each of the fuel cells that each provide a portion of the voltage to the load.

Fuel cell **210** may be used in any suitable fuel cell context, particularly a fuel cell context where dynamic load changes

may occur. One example application for fuel cell **210** is for use in a data center, as shown below in accordance with one example in FIG. 3.

Illustrative Data Center

FIG. 3 is a diagram of a portion (**305**) of an example data center. Portion **305** may include fuel cells **300**, batteries **389**, voltage regulators **381**, voltage regulators **382**, DC racks **378**, and AC racks **379**. In some examples, data center services are provided by server racks including DC racks **378** and AC racks **379**, where the fuel cells **300** are used as the power source for the server racks. Each fuel cell **300** has a corresponding battery **389** for fuel cell **300** in some examples. Voltage regulators **381** are used at the output of batteries **389** in some examples.

In some examples, the output of each fuel cell **300** goes to a corresponding voltage regulator **382**, to provide a constant voltage output responsive to the received voltage output by the corresponding fuel cell **300**. In some examples, the output of voltage regulators **382** are used to power DC racks **378** and AC racks **379**, where each of the AC racks include a DC/AC converter. In the examples illustrated, a common fuel line comes in to provide gaseous fuel to each of the fuel cells **300**. The gaseous fuel may be hydrogen gas, carbon monoxide gas, a fuel source that will react in fuel cell **300** to form a suitable fuel in the hot box of the fuel cell **300**, or the like. In some examples, the server racks are powered on a row-by-row basis, with a row of server racks connected to a row of fuel cells. Other suitable arrangements are used in other examples.

FIG. 3 shows a particular example of a portion of an example data center. Various other examples of the disclosure may include more or less components than shown, and may vary according to various suitable manners.

Illustrative Computing Device

FIG. 4 is a diagram illustrating one example of computing device **400** in which aspects of the technology may be practiced. Computing device **400** may be virtually any type of general- or specific-purpose computing device. In some examples, computing device **400** may be an example of a portion of control subsystem **220** of FIG. 2 and or control subsystem **120** of FIG. 1. As illustrated in FIG. 4, computing device **400** includes processing circuit **410**, operating memory **420**, memory controller **430**, data storage memory **450**, input interface **460**, output interface **470**, and network adapter **480**. Each of these afore-listed components of computing device **400** includes at least one hardware element.

Computing device **400** includes at least one processing circuit **410** configured to execute instructions, such as instructions for implementing the herein-described workloads, processes, or technology. Processing circuit **410** may include a microprocessor, a microcontroller, a graphics processor, a coprocessor, a field-programmable gate array, a programmable logic device, a signal processor, or any other circuit suitable for processing data. Processing circuit **410** is an example of a core. The aforementioned instructions, along with other data (e.g., datasets, metadata, operating system instructions, etc.), may be stored in operating memory **420** during run-time of computing device **400**. Operating memory **420** may also include any of a variety of data storage devices/components, such as volatile memories, semi-volatile memories, random access memories, static memories, caches, buffers, or other media used to store run-time information. In one example, operating memory **420** does not retain information when computing device **400** is powered off. Rather, computing device **400** may be configured to transfer instructions from a non-volatile data

storage component (e.g., data storage component **450**) to operating memory **420** as part of a booting or other loading process.

Operating memory **420** may include 4th generation double data rate (DDR4) memory, 3rd generation double data rate (DDR3) memory, other dynamic random-access memory (DRAM), High Bandwidth Memory (HBM), Hybrid Memory Cube memory, 3D-stacked memory, static random-access memory (SRAM), or other memory, and such memory may comprise one or more memory circuits integrated onto a DIMM, SIMM, SODIMM, or other packaging. Such operating memory modules or devices may be organized according to channels, ranks, and banks. For example, operating memory devices may be coupled to processing circuit **410** via memory controller **430** in channels. One example of computing device **400** may include one or two DIMMs per channel, with one or two ranks per channel. Operating memory within a rank may operate with a shared clock, and shared address and command bus. Also, an operating memory device may be organized into several banks where a bank can be thought of as an array addressed by row and column. Based on such an organization of operating memory, physical addresses within the operating memory may be referred to by a tuple of channel, rank, bank, row, and column.

Despite the above-discussion, operating memory **420** specifically does not include or encompass communications media, any communications medium, or any signals per se.

Memory controller **430** is configured to interface processing circuit **410** to operating memory **420**. For example, memory controller **430** may be configured to interface commands, addresses, and data between operating memory **420** and processing circuit **410**. Memory controller **430** may also be configured to abstract or otherwise manage certain aspects of memory management from or for processing circuit **410**. Although memory controller **430** is illustrated as single memory controller separate from processing circuit **410**, in other examples, multiple memory controllers may be employed, memory controller(s) may be integrated with operating memory **420**, or the like. Further, memory controller(s) may be integrated into processing circuit **410**. These and other variations are possible.

In computing device **400**, data storage memory **450**, input interface **460**, output interface **470**, and network adapter **480** are interfaced to processing circuit **410** by bus **440**. Although, FIG. 4 illustrates bus **440** as a single passive bus, other configurations, such as a collection of buses, a collection of point to point links, an input/output controller, a bridge, other interface circuitry, or any collection thereof may also be suitably employed for interfacing data storage memory **450**, input interface **460**, output interface **470**, or network adapter **480** to processing circuit **410**.

In computing device **400**, data storage memory **450** is employed for long-term non-volatile data storage. Data storage memory **450** may include any of a variety of non-volatile data storage devices/components, such as non-volatile memories, disks, disk drives, hard drives, solid-state drives, or any other media that can be used for the non-volatile storage of information. However, data storage memory **450** specifically does not include or encompass communications media, any communications medium, or any signals per se. In contrast to operating memory **420**, data storage memory **450** is employed by computing device **400** for non-volatile long-term data storage, instead of for run-time data storage.

Also, computing device **400** may include or be coupled to any type of processor-readable media such as processor-

readable storage media (e.g., operating memory **420** and data storage memory **450**) and communication media (e.g., communication signals and radio waves). While the term processor-readable storage media includes operating memory **420** and data storage memory **450**, the term “processor-readable storage media” (whether in the plural or singular form), throughout the specification and the claims, is defined herein so that the term “processor-readable storage media” specifically excludes and does not encompass communications media, any communications medium, or any signals per se. However, the term “processor-readable storage media” does encompass processor cache, Random Access Memory (RAM), register memory, and/or the like.

Computing device **400** also includes input interface **460**, which may be configured to enable computing device **400** to receive input from users or from other devices. In addition, computing device **400** includes output interface **470**, which may be configured to provide output from computing device **400**. In one example, output interface **470** includes a frame buffer, graphics processor, graphics processor or accelerator, and is configured to render displays for presentation on a separate visual display device (such as a monitor, projector, virtual computing client computer, etc.). In another example, output interface **470** includes a visual display device and is configured to render and present displays for viewing. In yet another example, input interface **460** and/or output interface **470** may include a universal asynchronous receiver/transmitter (“UART”), a Serial Peripheral Interface (“SPI”), Inter-Integrated Circuit (“I2C”), a General-purpose input/output (GPIO), and/or the like. Moreover, input interface **460** and/or output interface **470** may include or be interfaced to any number or type of peripherals.

In the illustrated example, computing device **400** is configured to communicate with other computing devices or entities via network adapter **480**. Network adapter **480** may include a wired network adapter, e.g., an Ethernet adapter, a Token Ring adapter, or a Digital Subscriber Line (DSL) adapter. Network adapter **480** may also include a wireless network adapter, for example, a Wi-Fi adapter, a Bluetooth adapter, a ZigBee adapter, a Long-Term Evolution (LTE) adapter, or a 5G adapter.

Although computing device **400** is illustrated with certain components configured in a particular arrangement, these components and arrangement are merely one example of a computing device in which the technology may be employed. In other examples, data storage memory **450**, input interface **460**, output interface **470**, or network adapter **480** may be directly coupled to processing circuit **410**, or be coupled to processing circuit **410** via an input/output controller, a bridge, or other interface circuitry. Other variations of the technology are possible.

Some examples of computing device **400** include at least one memory (e.g., operating memory **420**) adapted to store run-time data and at least one processor (e.g., processing unit **410**) that is adapted to execute processor-executable code that, in response to execution, enables computing device **400** to perform actions.

Illustrative Process

FIGS. 5A-5B illustrate an example dataflow for a process (**590**) for a fuel cell. In some examples, process **590** is performed by a device, such as fuel cell **100** of FIG. 1, fuel cell **200** of FIG. 2, or one or more of the fuel cells **300** of FIG. 3.

In the illustrated example, step **591** occurs first. At step **591**, in some examples, a load is caused to be driven based, at least in part, on a voltage difference between an anode and a cathode of a fuel cell stack of a fuel cell. As shown, step

592 occurs next in some examples. At step **592**, in some examples, at least one characteristic associated with the load is measured. As shown, step **593** occurs next in some examples. At step **593**, in some examples, at least one control signal is provided based, at least in part, on the at least one characteristic.

As shown, step **594** occurs next in some examples. At step **594**, in some examples, an oxidizing agent is provided to the cathode of the fuel cell stack. As shown, step **595** occurs next in some examples. At step **595**, in some examples, gaseous fuel is pumped into the anode of the fuel cell stack. As shown, step **596** occurs next in some examples. At step **596**, in some examples, mass flow of the gaseous fuel to the cathode of the fuel cell stack is adjusted based on at least a first control signal of the at least one control signal. The process may then advance to the return block, where other processing is resumed.

CONCLUSION

While the above Detailed Description describes certain examples of the technology, and describes the best mode contemplated, no matter how detailed the above appears in text, the technology can be practiced in many ways. Details may vary in implementation, while still being encompassed by the technology described herein. As noted above, particular terminology used when describing certain features or aspects of the technology should not be taken to imply that the terminology is being redefined herein to be restricted to any specific characteristics, features, or aspects with which that terminology is associated. In general, the terms used in the following claims should not be construed to limit the technology to the specific examples disclosed herein, unless the Detailed Description explicitly defines such terms. Accordingly, the actual scope of the technology encompasses not only the disclosed examples, but also all equivalent ways of practicing or implementing the technology.

We claim:

1. An apparatus, comprising:

at least one memory storing processor-executable code therein, and at least one processor that is adapted to execute the processor-executable code, wherein the processor-executable code includes processor-executable instructions that, in response to execution, enables the apparatus to perform actions, including:

measuring at least one characteristic associated with a load that is driven based, at least in part, on a voltage difference between an anode and a cathode of a fuel cell stack of a fuel cell, wherein the at least one characteristic associated with the load is at least one of a voltage of the load or a current of the load;

controlling pumping of gaseous fuel into the fuel cell; providing a plurality of control signals based, at least in part, on the at least one characteristic, such that a first control signal causes an adjustment of mass flow of the gaseous fuel to the anode of the fuel cell stack and such that a second control signal causes control of the pumping of the gaseous fuel into the fuel cell;

controlling pumping of an oxidizing agent into the cathode of the fuel cell stack based on a third control signal; adjusting a speed of the pumping of the oxidizing agent based on a fourth control signal; and

adjusting a speed of the pumping of the gaseous fuel into the fuel cell based on a fifth control signal,

wherein the speed of the pumping of the oxidizing agent and the speed of the pumping of the gaseous fuel are linked.

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2. The apparatus of claim 1, the actions further including: storing the gaseous fuel; and providing the stored gaseous fuel to the fuel cell stack responsive to a drop in pressure in the gaseous fuel being pumped into the fuel cell.

3. The apparatus of claim 1, the actions further including: causing an adjustment of a speed of the pumping of the gaseous fuel into the fuel cell stack based, at least in part, on the second control signal.

4. The apparatus of claim 1, the actions further including: providing an inert gas to the anode of the fuel cell stack responsive to a sixth control signal.

5. The apparatus of claim 1, the actions further including: using a hot box that surrounds the fuel cell stack to insulate components outside of the hot box from high temperature inside the hot box.

6. The apparatus of claim 4, the actions further including adjusting a mass flow of the oxidizing agent into the cathode of the fuel cell stack based on a seventh control signal.

7. A method, comprising:

measuring at least one characteristic associated with a load that is driven based, at least in part, on a voltage difference between an anode and a cathode of a fuel cell stack of a fuel cell, wherein the at least one characteristic associated with the load is at least one of a voltage of the load or a current of the load;

controlling pumping of gaseous fuel into the fuel cell; and providing a plurality of control signals based, at least in part, on the at least one characteristic, such that a first control signal causes an adjustment of mass flow of the gaseous fuel to the anode of the fuel cell stack and such that a second control signal causes control of the pumping of the gaseous fuel into the fuel cell;

controlling pumping of an oxidizing agent into the cathode of the fuel cell stack based on a third control signal; adjusting a speed of the pumping of the oxidizing agent based on a fourth control signal;

adjusting a speed of the pumping of the gaseous fuel into the fuel cell based on a fifth control signal, and wherein the speed of the pumping of the oxidizing agent and the speed of the pumping of the gaseous fuel are linked.

8. The method of claim 7, further comprising: storing the gaseous fuel; and providing the stored gaseous fuel to the fuel cell stack responsive to a drop in pressure in the gaseous fuel being pumped into the fuel cell.

9. The method of claim 7, further comprising: providing an inert gas to the anode of the fuel cell stack responsive to a sixth control signal.

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10. A processor-readable storage medium, having stored thereon processor-executable code that, upon execution by at least one processor, enables actions, comprising:

measuring at least one characteristic associated with a load that is driven based, at least in part, on a voltage difference between an anode and a cathode of a fuel cell stack of a fuel cell, wherein the at least one characteristic associated with the load is at least one of a voltage of the load or a current of the load;

controlling pumping of gaseous fuel into the fuel cell; and providing a plurality of control signals based, at least in part, on the at least one characteristic, such that a first control signal causes an adjustment of mass flow of the gaseous fuel to the anode of the fuel cell stack and such that a second control signal causes control of the pumping of the gaseous fuel into the fuel cell;

controlling pumping of an oxidizing agent into the cathode of the fuel cell stack based on a third control signal; adjusting a speed of the pumping of the oxidizing agent based on a fourth control signal;

adjusting a speed of the pumping of the gaseous fuel into the fuel cell based on a fifth control signal, and wherein the speed of the pumping of the oxidizing agent and the speed of the pumping of the gaseous fuel are linked.

11. The processor-readable storage medium of claim 10, the actions further comprising: causing linking an adjustment of mass flow of an oxidizing agent to a cathode of the fuel cell stack with the adjustment of the mass flow of the gaseous fuel to the anode of the fuel cell stack.

12. The processor-readable storage medium of claim 10, the actions further comprising: storing the gaseous fuel; and providing the stored gaseous fuel to the fuel cell stack responsive to a drop in pressure in the gaseous fuel being pumped into the fuel cell.

13. The processor-readable storage medium of claim 10, the actions further comprising: providing an inert gas to the anode of the fuel cell stack responsive to a sixth control signal.

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